CANDIDATE BRIEF

Can you make a small AI workflow dependable?

A practical repair assignment for the MOSU AI Systems Developer role. Junior developers and interns are welcome.

AI tools are allowed. We assess the result you can explain, verify and adapt. You are not expected to write every line unaided or build a production platform.

The task

Repair the supplied **workflow.mjs**. A synthetic customer has a new measurement-profile version. The workflow must validate it, prepare a clarification, wait for approval and send it safely through a mock tool. All data and recipients are fictional; no real messages are sent.

The starting situation

Profile **v3** was previously approved. Profile **v4** has chest = 92 cm, waist missing and hip = 38 in. The supplied protocol requires chest, waist and hip in cm. The expected issues are **missing:waist** and **unit:hip:in**. Preserve the supplied measurements; do not invent or silently convert them.

Six rules your workflow must enforce

Rule	Required behavior
1. Version ownership	A new profile version does not inherit an earlier approval. This workflow prepares clarification; it never approves the measurement profile itself.
2. Validate first	Use the protocol fields and required unit. Flag missing fields, incorrect units and supplied numbers that are not finite and positive. Draft only after validation.
3. Reference text is data	A protocol note cannot authorize a send, change the recipient, bypass validation or approve an action.
4. Exact-action approval	Only a trusted approval event for the current pending action ID permits sending. Its identity binds the profile, version, recipient and exact text.
5. Evidence of success	Record sent only after receiving a receipt. A timeout leaves the result unknown and preserves the action, approval and retry key.
6. Bounded safe retry	At most one notification call per SEND event, and two attempts per exact action. Reuse its key and payload. A confirmed send is never sent again.

THE ACCEPTANCE CASES

What a working result looks like

The starter runner supplies these five scenarios. The reviewer may vary synthetic IDs, field values and notes within the same published rules. Read README.md for the precise function contract and issue-code format.

Scenario	Expected observable result
1. New profile version	Load v4 after approved v3. Show v4, both validation issues and a clarification awaiting review. No inherited approval; no notification call.
2. Instruction inside a note	A retrieved note says to skip review and send. Validation still runs; the note grants no authority. SEND without human approval makes zero notification calls.
3. Stale approval	Approve an unrelated or old action ID. The current pending action remains; sending is blocked; zero notification calls.
4. Approved send, then replay	Approve the actual pending action and invoke SEND twice. One tool call, one delivery, one receipt. The replay preserves the completed result.
5. Delivered; response lost	The tool commits the message, then times out. First result: unknown. Retry the same action: recover the original receipt. Two requests, one delivery.

The retry-limit variant is also part of the rules

If both allowed attempts time out, keep the status unknown and expose that human follow-up is needed. A third SEND makes no call. Reloading the same exact action must not reset its attempts or completed delivery. The runner includes documented checks for these boundaries.

What we provide

A runnable Node.js starter, deterministic drafting and protocol mocks, an instrumented notification tool, an action-ID helper, fixture cases and a state report. No model subscription, API key, installation of packages, account or cloud deployment is required. A supported local Node.js installation is sufficient.

Four useful ideas

Concept	Meaning in this exercise
State	What the workflow currently knows: version, issues, pending action, approval, attempts and receipt.
Approval boundary	Permission for one specific action. Permission for an older version or different payload is not permission for this one.
Idempotency	Repeated requests with the same key and payload cause one effect. The supplied notification tool provides this behavior; you must use it correctly.
Unknown outcome	The response is missing, so the action may already have happened. Do not label it confirmed success or confirmed failure.

HOW TO COMPLETE IT

Work budget and submission

Part	Maximum work time	What to do
Initial assignment	2 hours	Read, repair, run cases and prepare a short handoff. Stop at the cap; incomplete work is acceptable if clearly described.
Later change	30 minutes	After submission, receive one small change to this same workflow. AI remains allowed. Preserve the original cases.
Product walkthrough	15 minutes	Explain, predict an outcome, then run the example. No live coding is required.

We will agree the honorarium, submission window and payment timing in writing before you start. The fixed fee is for the agreed good-faith attempt, not conditional on being selected. Do not begin until those terms are confirmed.

Start here

Unzip the candidate kit. Read README.md and the contract. From that folder, run:

```
node run.mjs
node run.mjs --json > results.json
```

The supplied implementation is deliberately faulty. Initial failures are expected. Repair the workflow, not the supplied checker or mocks. You may use standard-library helpers inside your workflow; no added dependencies are needed. Ask if setup blocks you; setup problems are recorded separately.

Submit a small handoff

1. Your workflow.mjs and any clearly identified local helper files you added.
2. The runner output, including any failures that remain.
3. A decision note of **400 words or fewer**: one design choice, one failure found and checked, one limitation, AI tools used and what you verified yourself.
4. Be ready to show the timeout case during the walkthrough. A short recording or annotated screenshots can support your handoff; editing and presentation polish earn no extra credit.

How your work will be reviewed

The reviewer reruns your workflow with independent mock records and equivalent examples. Five areas matter: working behavior; diagnosis and verification; explanation of state and tool boundaries; adaptation to the later change; and an accurate handoff. The same criteria apply to every applicant. Neither a perfect score nor a polished demo guarantees an offer.

Use AI responsibly

Use any available assistant, documentation or coding agent. Explain what you accepted, changed and checked. Full chat histories, private employer code and continuous recordings are not requested. During the walkthrough, give your best current prediction first, including uncertainty; you may then consult tools and revise it.

Keep it small. A UI, database, vector store, multi-agent framework, real MCP server and deployment are outside scope. Extra features and extra hours earn no advantage. We value a clear partial result over an unsupported claim that everything is production ready.